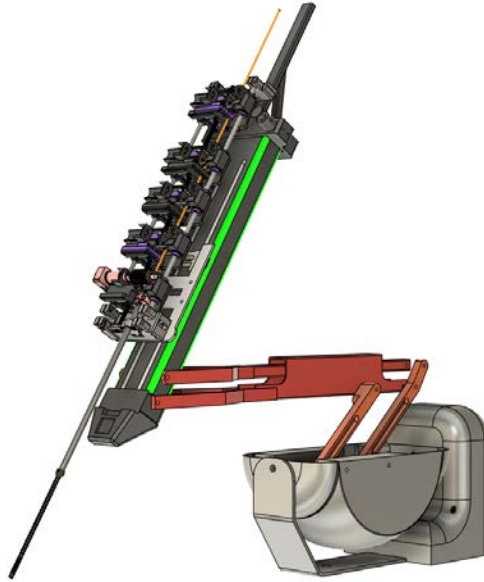
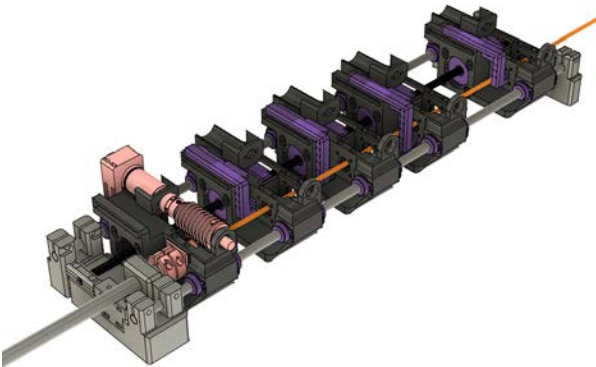
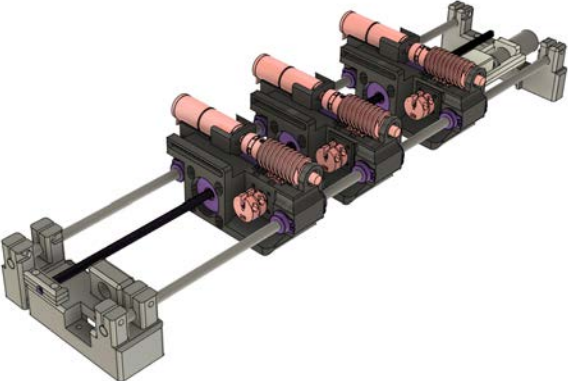


Module Assembly Instructions:

Modular Multi-Continuum Robot Platform for the da Vinci Research Kit (dVRK)



Mechanical System

Tendon-driven CR (TDCR Actuation System Assembly: (excluding CR)	Concentric Tube Robot Actuation System Assembly: (excluding CR)
	
• 1 roll module ○ For roll module assembly do only the following: ■ Worm Assembly (Bottom, Internal Section)	• 3 full modules ○ For roll module assembly do only the following: ■ Worm Assembly (Bottom, Internal Section)

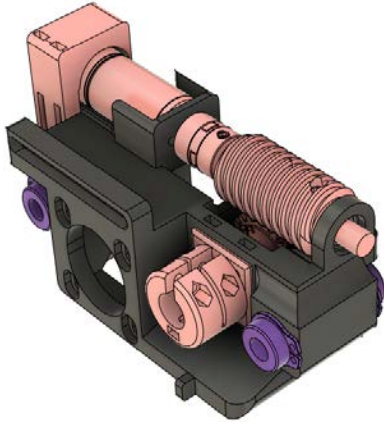
- **Worm Coupling Assembly (Top Section)**

- 4 tendon modules
 - For tendon module assembly do only the following:
 - **Linear Bearings for Motion Shafts**
 - **Stepper Motor**
 - **Limit Switch**
 - **Wire Router**

- **Worm Coupling Assembly (Top Section)**
- **Linear Bearings for Motion Shafts**
- **Stepper Motor**
- **Limit Switch**
- **Wire Router**

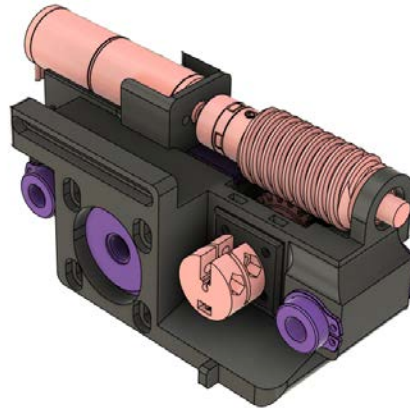
1x roll module

- Rotation subassembly
- Missing translation assembly



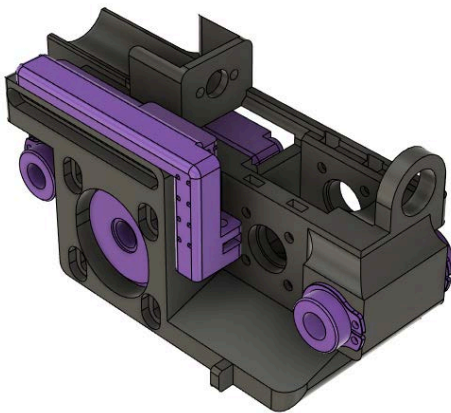
3x full modules

- Rotation subassembly
- Translation subassembly



1x tendon module

- Missing rotation subassembly
- Translation assembly



For manufacturing of all custom components, outside of the tools mentioned, you require access to:

- 3D Printer FDM - i.e. Bambu Lab X1 Carbon
 - 3D Printer SLA - i.e. Formlabs Form 4 printer
 - Metal Laser Cutter - Fablight
 - Hand saw OR band saw
-

Module Component List:

NOTE: Item quantities mentioned are for 1 module with both, the translational and rotational mechanisms

1x Module Frame



2x Hollow Shaft Bearing (Closed Housing)



1x Worm Wheel



1x Worm



1x Hollow Shaft



1x Worm Shaft



1x Brushless Direct Current (BLDC) Motor



1x BLDC Clamp



1x Motor Shaft Coupling 	1x Coupling Disc 
1x Worm Shaft Coupling 	1x Worm Shaft Bearing (Open Housing) 
1x Worm-Bearing Spacer 	1x Backlash Panel 
2x Linear Bearing 	4x Snap Ring 
1x Stepper Motor	1x Wire Router



1x Limit Switch

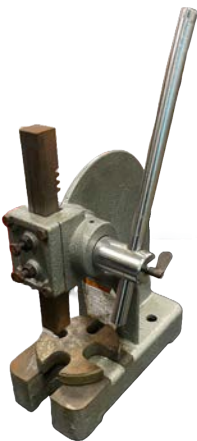


1x Tube Clamp



General Tools:

Arbor Press



Snap Ring Pliers



M2.5 Hex Key

M2 Hex Key



Bearing Assembly Box Tool



Flat-end Tweezers



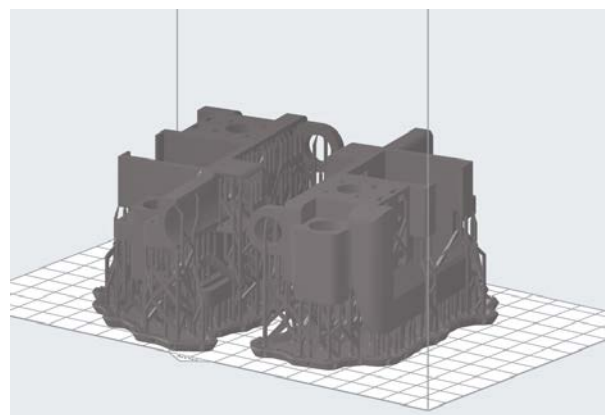
Manufacturing Instructions

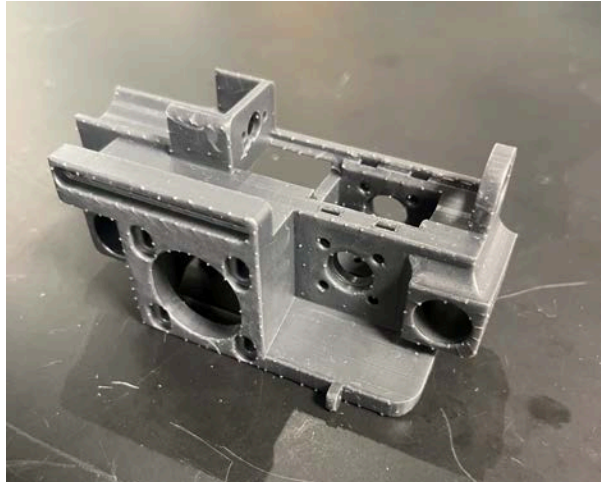
Module Frame (resin printing):

- Components Required
 - Form 4
 - Tough Resin 1500 V2

Print Options in Slicer:

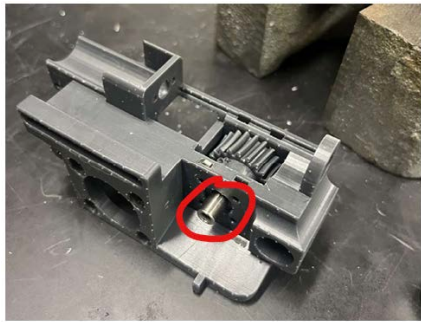
- Print in the orientation such that no supports in the linear bearing (make the front face the bottom)
- Change the touchpoint size of supports to 0.30 mm for better surface finish
- Preform file in Github repo available for download



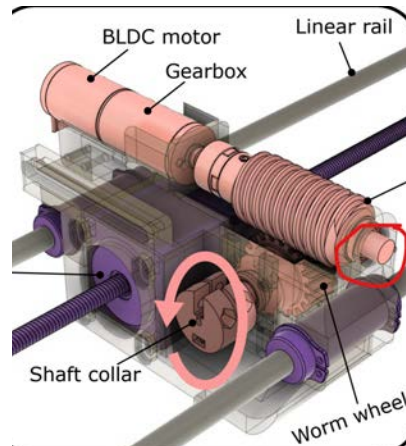
Post Processing: • Wash in IPA wash station for 15 minutes • Shake dry and drain IPA as much as possible • Cure in cure station for 60 minutes at 70 degrees Celsius	Standard Formlabs cleaning procedure
Support Clearing: • Use wire cutters to cut off supports and break off the rest by hand, ensure that the back face of the resin module where the slot for the worm wheel lies and edges of the linear bearing holders are flat, removing support remnants through sanding/scraping	

Shaft Cuts to Length:

- Components Required
 - Band saw / hand saw
 - Hollow shaft
 - Worm shaft
 - Lead screw

Cut hollow shaft using a band saw or hand saw to 36 mm (37 mm is also okay)	 Hollow shaft indicated above
--	--

Cut worm shaft using a band saw or hand saw to 47 mm (48 mm is also okay)



Worm shaft indicated above

Use a file and/or a belt sander to remove burrs and fix sections of the shafts bowing in or out

Cutting lead screw:

- To maintain the cleanness of the thread, put the hex nut deep onto the lead screw before cutting
- Cut the lead screw using a band saw or hand saw to 400 mm or to desired length.
- Unscrew hex nut to make the thread of the lead screw undamaged and clean out remnant chips

Module Assembly Instructions

NOTE: item quantities listed of components required are per module and not representative of how many are required for a complete system

Worm Wheel Assembly (Bottom Internal Section, Worm Wheel):

- Components Required
 - 2x hollow shaft bearing
 - 1x backlash panel
 - 1x worm wheel
 - 4x M2 hex nuts
 - 4x M2x6 bolts
- Auxiliary Items
 - Arbor press
 - M2 hex key

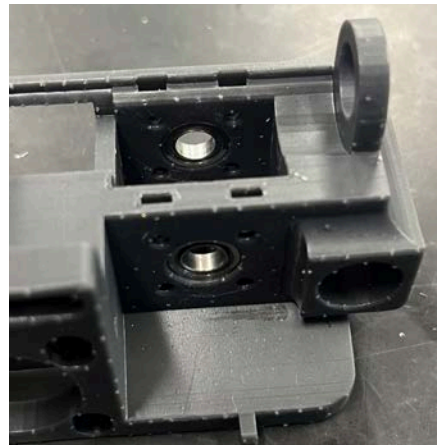
Put the hollow shaft bearing in the front wall housing with the flange closer to the front side, ensure it has snapped into place within its housing

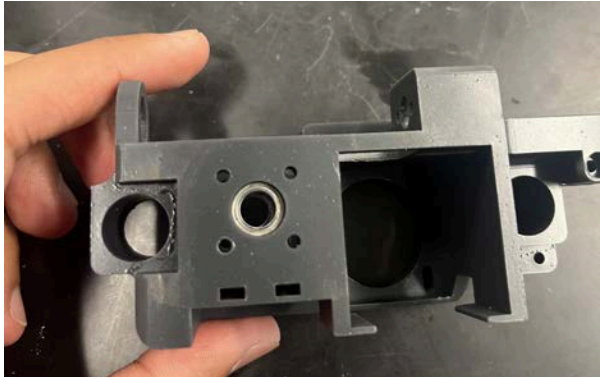
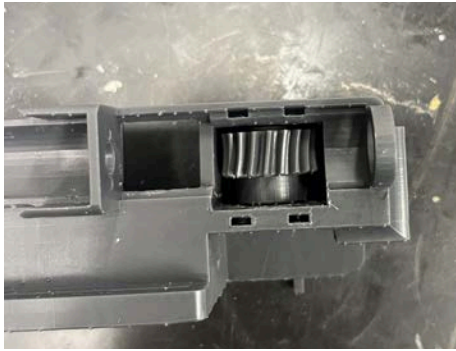
- Push as far as possible, an existing flange limits maximum depth to intended position

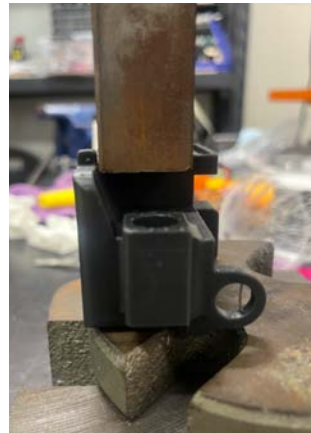


Fit the second hollow shaft bearing in the back wall into place with a snap, ensure the flange is closer to the front side

- Fit it from the inside hollow section and push towards the back, do not fit it from the outside face inwards
- Push as far as possible, an existing flange limits maximum depth to intended position



	
Worm wheel • Put M2 hex nuts into worm wheel • Place worm wheel inside module with gear teeth closer to the back wall of the module • Push the worm wheel inside until hole is somewhat concentric with bearings • Might be a very tight fit but it should work as intended	 
Exercise caution while using the arbor press here. • Ensure all sides of the square profile of the arbor press' ram fit within the module without scraping anything • Refer to pictures of what not to do: ○ In the first image, be careful of hitting the side wall and compressing or scraping that side ○ In the second image, be careful of hitting the sloped surface on the bottom of the module and scraping material	Correct alignment, what to do:  Be mindful of, what not to do:



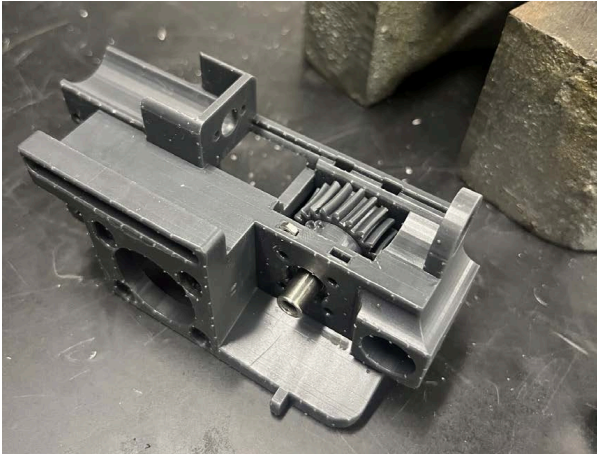
Press fit the hollow shaft into the first bearing using the bearing assembly tool

- Place module frame with bearings inside and worm wheel inside with back face down in the arbor press
- Fit the bearing box assembly tool on top of the bearing and put the hollow shaft through it to ensure straight alignment when putting the hollow shaft in
- Once shaft is flush with the top of the bearing box assembly tool, the shaft is successfully press fit into the first bearing, you have successfully put the bearing box assembly tool away



Continue the press fit through, push through the worm wheel and then the bearing at the back as well

Final product of press fitting through all components:

• Ensure that the back of module is against a flat face so that you can press fit all the way through to the back • If you observe flexing of the module, you are applying too much force • If it feels stuck, release the arbor press force applied and reapply it • The back wall bearing may require more force to self-align and enter	
Secure the hollow shaft bearing in the front wall with a backlash panel • Begins by putting M2 hex nuts into top slots and pushing hex nuts to full depth and then flipping module frame and putting M2 hex nuts into bottom slots by pushing hex nuts to full depth • Screw on backlash panel using 2 M2x6 bolts in the diagonal holes for alignment • Put M2 hex nuts into the remaining two slots and push hex nuts fully in • Secure bottom section of backlash panel using another 2 M2x6 bolts	
For disassembly: use M5 or M6 bolt and push inner hollow shaft out using an arbor press	

Worm Gear Assembly (Top Section, Worm)

- Components Required
 - 1x BLDC motor
 - 1x BLDC clamp
 - 1x motor shaft coupling
 - 1x coupling disc
 - 1x worm shaft coupling
 - 1x worm shaft bearing
 - 1x worm

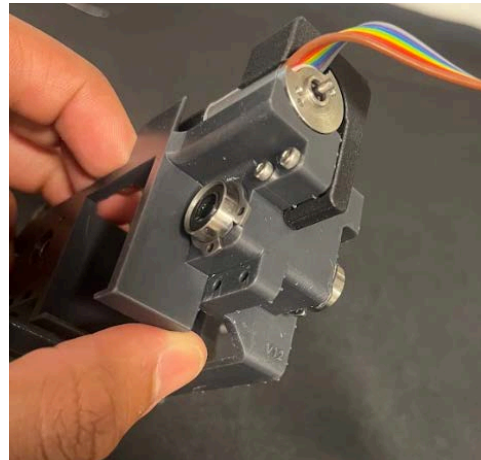
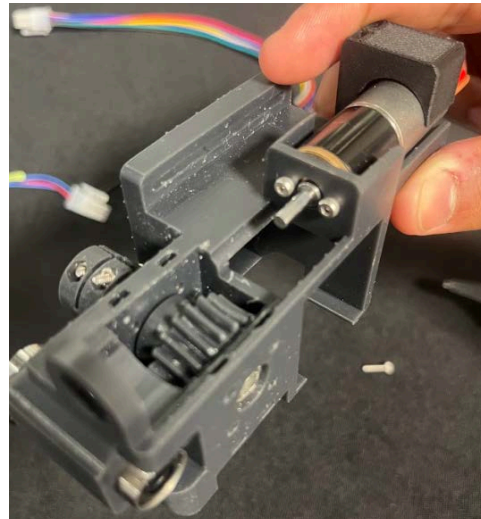
- 1x worm wheel
 - 2x M2 hex nuts
 - 2x M2x8 screws
 - 2x M2x12 screws
- Auxiliary Items
 - M2 Hex Key
 - M2.5 Hex Key

Screw in the BLDC motor with shaft sticking out as shown in image

- Use M2x8 screws

Use the BLDC clamp to hold the back of the BLDC motor in place

- Put 2 M2 hex nuts into the vertical hex nut slots in the BLDC clamp
- Put the BLDC clamp closer to the front side of the module
- Screw in 2 M2x12 bolts through the module, clamp, and hex nut to tighten it at the intended height just touching the BLDC motor



Put the motor shaft coupling, coupling disc, and worm shaft coupling together, they fit together in a snap fit, orientation relative to one another does not matter as long as it fits

- Motor shaft coupling is the set screw piece with the smaller inner diameter
- Worm shaft coupling is the set screw piece with the bigger inner diameter
- Coupling disc is the thin, 3D-printed piece in between

Attach the motor shaft coupling (motor shaft part) using its built-in set screw onto the flat edge of the motor shaft (D shaft flat edge)

- The motor shaft has a flat face, the set screw should tighten onto this flat face to ensure concentricity

Ensure that the worm shaft coupling (set screw piece on the other side is loosened)

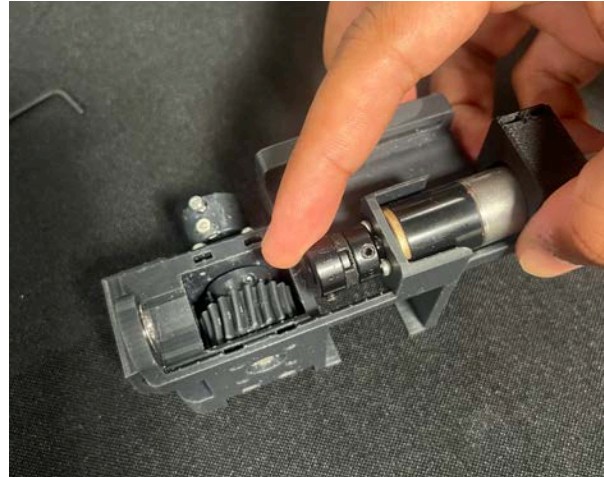


Put the worm shaft bearing into its housing on the inside face of the module

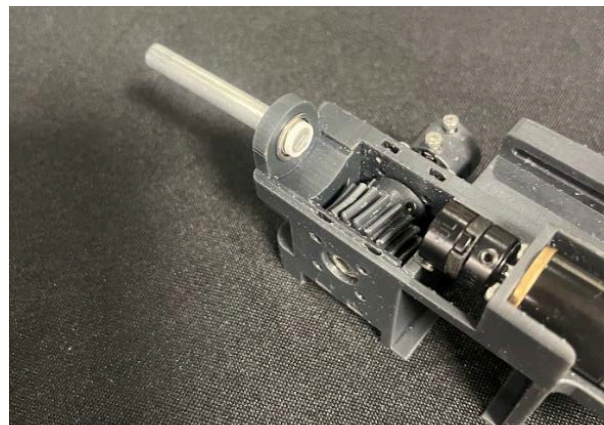
- The flange is on the inside of the housing

Loosen the motor shaft coupling and push it all the way back towards the motor as shown in the image





Hold the 3D-printed worm spacer in place and slide the worm shaft through the bearing and spacer as shown in the image



Put the worm into its place between the spacer and the couplings

- The worm should be placed such that the built-in set screw is on the side that is closer to the worm shaft bearing than the BLDC motor



Push the actual worm shaft into the bearing and all the way through the worm spacer, into the worm, and into the worm shaft coupling

Loosen the motor shaft coupling and push the coupling and the rest of the sub assembly in the direction of the bearing such that the whole assembly is in contact, as shown in picture

While holding as shown in picture, tighten all set screws



Re-tighten motor shaft coupling and tighten other set screws (worm shaft couple, worm)



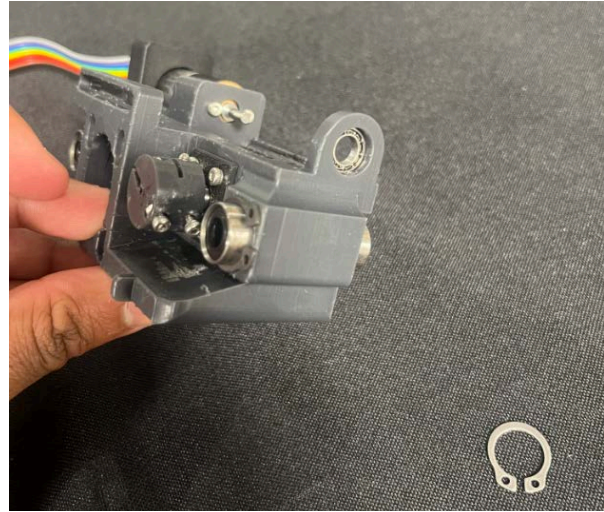
Linear Bearings for Motion Shafts

- Components Required
 - 2x linear bearings
 - 4x snap rings
- Auxiliary Items
 - Snap ring pliers

Exercise caution when handling snap rings with snap ring pliers, be extremely careful of opening the snap rings too far, they have a tendency to permanently (plastically) deform after which they are unusable

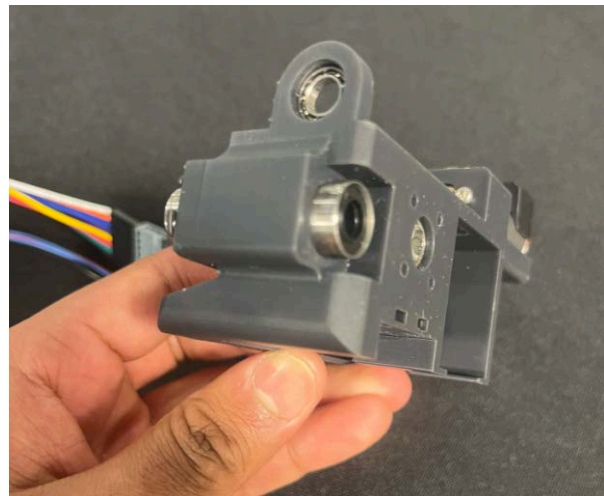


Put the snap ring on one side of the linear bearing inside the groove



Push the linear bearings into the circular slots on the sides of the module for it until the inside face of the snap ring hits the module

- Might need arbor press to push linear bearings in if very tight



Put other snap ring on other side of the linear bearing in the groove (should be in opposite orientation as the other snap ring)



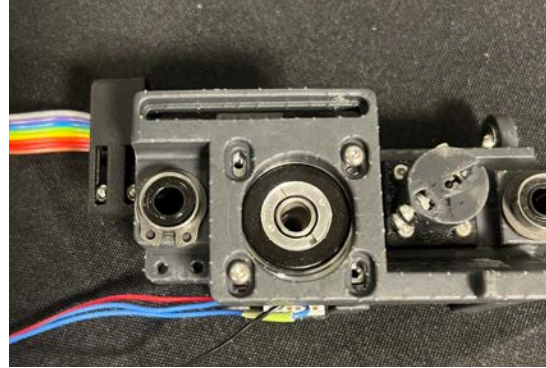
Stepper Motor

- Components Required
 - 1x stepper motor

- 4x M2.5x8 screws
- Auxiliary Items
 - M2.5 hex key

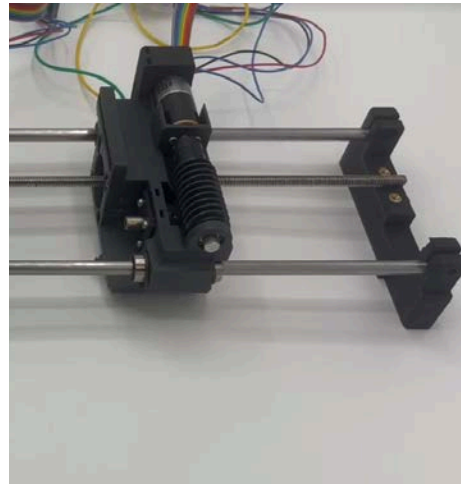
Place stepper motor into the slot in the module for it with electrical connector at the bottom facing forward

- Loosely tighten the M2.5x8 screws into the stepper motor (at least 2 diagonal)



When system frame is assembled with the lead screw and linear motion shafts, put the module onto the lead screw and linear motion shafts and let the stepper motor naturally settle into its appropriate vertical position

- Ensures alignment of stepper motor on module frame relative to the alignment of lead screw and linear motion shafts

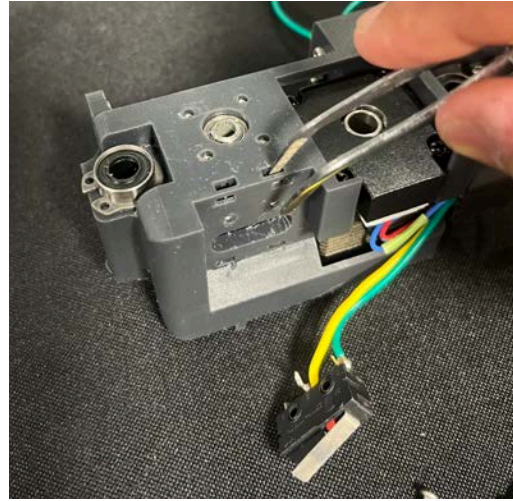


Limit Switch

- Components Required
 - 1x limit switch
 - 2x M2 hex nuts
 - 2x M2x10 screws
- Auxiliary Items
 - M2 hex key

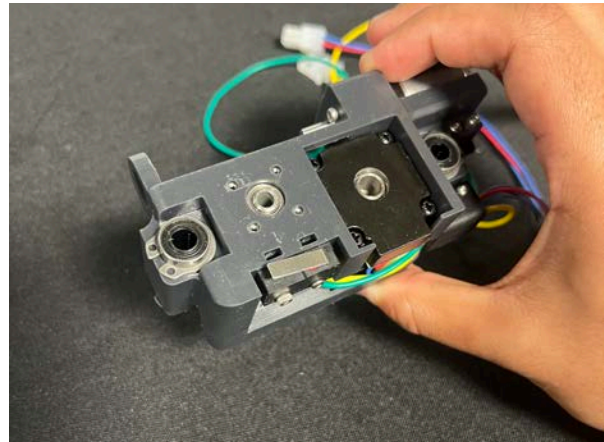
Put the hex nut into the hex nut slots (M2)

- Ensure they are pushed into full depth
- Refer to image, insert hex nuts into slots where tweezers are entering

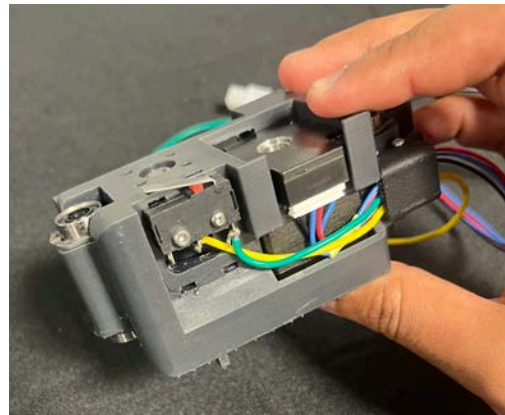


Hold the limit switch holes in line with the bolt holes and make sure the pressable tab faces the back of the module

- When looking at the back face of the module, the part that sticks out (pressable tab) more should be on the left side



Use M2x10 bolts to screw limit switch into place

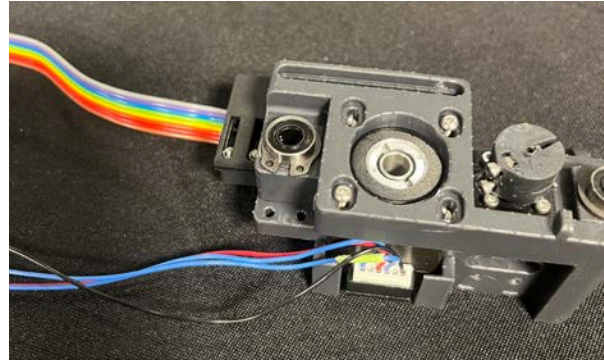


Wire Router

- Components Required
 - 1x wire router

- 2x M2 hex nuts
- 2x M2x16 screws
- Auxiliary Items
 - M2 hex key

Keep stepper motor electrical wires and limit switch electrical wires close to the bottom face of the module and route it outwards from the left side (when looking at the front face of the module)

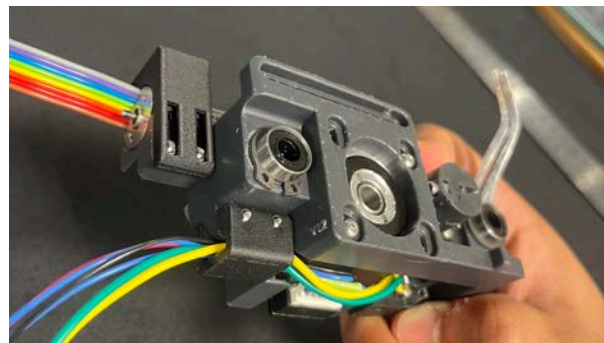


Put hex nuts inside the wire router part

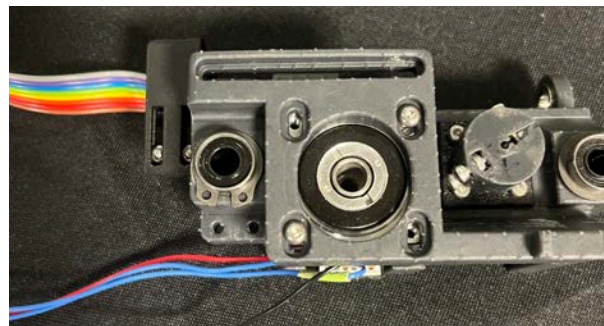
- Be careful and keep vertical orientation of this part during rest of assembly to prevent hex nuts from falling



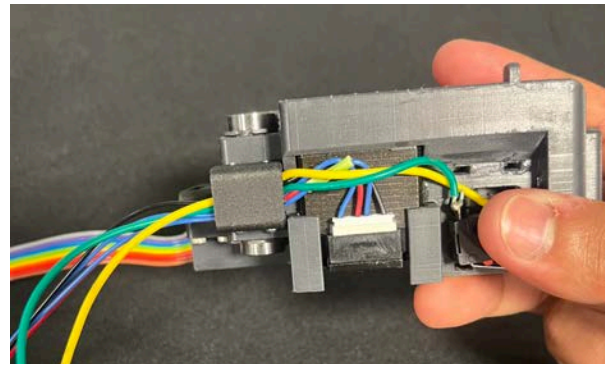
Slot the wires into the wire router part snugly



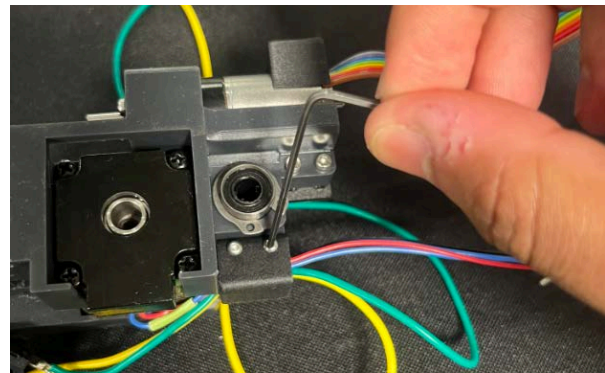
Take the wire router part (with wires in place) and put it such that the thicker part is closer to the front face of the module



Ensure all 6 wires are inside in the gap left by the wire router in between the module

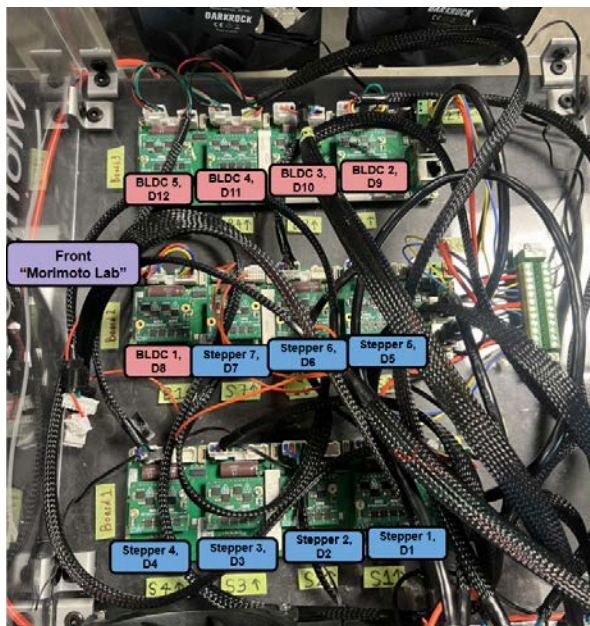


Screw the wire router part on using a M2x16 screw



Electronics Assembly Instructions

NOTE: item quantities listed of components required are per module and not representative of how many are required for a complete system

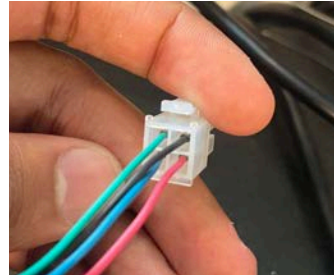


Molex Connections:

- Components Required
 - 4-pin Molex connectors, male and female
 - 8-pin Molex connectors, male and female
 - Crimp pins, male and female
 - 8 conductor wire
 - 4 conductor wire
 - Molex crimper or standard crimper

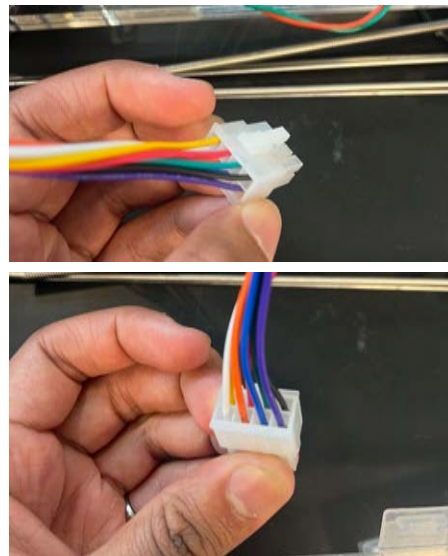
Stepper Motor to Molex Tutorial

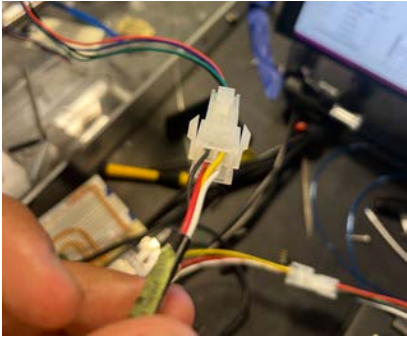
- Strip 0.5 cm off of each wire (already done for steppers)
- Get female Molex crimp pin and male Molex connectors
- Crimp each individual wire to the crimp terminal using the appropriate gauge setting
 - Ensure outer tab is on top of rubber external and inner tab is on top of wire core
- Slide female crimped wire into male crimped connector
 - Black - 1, Green - 2, Red - 3, Blue - 4 (refer to picture)



BLDC Motor to Molex Tutorial

- Insert the pre-provided male crimped wire into the 8 pin connector
 - From Amazon order comes together
- Cut the wires on the other side and strip 0.5 cm of the wire
- Using the female Molex crimp pins, crimp them onto the stripped jumper cables that come off the BLDC motor
- Insert the female Molex crimp pins into the male Molex connector
 - Refer to picture for colour order



Molex to Long Wire (extends to dVRK) • Cut the long 22 gauge cables to 3 m (each motor should have one of these) • Strip the outer black wire about 4-5 cm • Strip 0.5 cm off of the internal wires • Crimp the stripped ends to a male Molex crimp pins • Slide those male Molex crimp pins into a female Molex connector ◦ This will connect to the male Molex connectors of the BLDC/Stepper ◦ Molex connectors are different for BLDC vs Steppers (8 pin vs 4 pin) • Repeat this process on the other end with the male Molex equivalent ◦ Female crimp pins and male Molex connectors which connect to the female Molex connectors on the IO board	Stepper long wire connections  BLDC long wire connections  
• For IO board-side connections refer to pictures	BLDC long wires, board-side:



Stepper long wires, board-side:

